Introduction to Artificial Intelligence

Sample learning document for testing a Retrieval-Augmented Generation (RAG) assistant

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a field of computer science focused on creating systems that can perform tasks that usually require human intelligence. These tasks include understanding language, recognizing images, solving problems, making predictions, and supporting decisions. An AI system receives input, processes information using a model or rules, and produces an output.

AI is used in everyday applications. Email services filter spam, navigation applications estimate travel times, and online stores recommend products based on browsing and purchase patterns. Hospitals may use AI tools to help analyze medical images, although important clinical decisions require qualified professionals.

2. Machine Learning

Machine Learning (ML) is a branch of AI in which systems learn patterns from data instead of relying only on explicitly written rules. A common workflow includes collecting data, cleaning it, selecting useful features, training a model, evaluating performance, and deploying the model.

Type	Description	Example
Supervised	Learns from labelled examples with expected answers.	Predicting house prices from past sales.
Unsupervised	Finds patterns or groups without labelled answers.	Grouping customers by shopping behaviour.
Reinforcement	Learns actions using rewards and penalties.	An agent learning to navigate a maze.

3. Deep Learning and Neural Networks

Deep Learning is a subfield of machine learning that uses neural networks with multiple layers. A neural network contains connected units called neurons. During training, the network adjusts connection weights to reduce the difference between predictions and correct answers. Neural networks are used in image classification, speech recognition, translation, and text generation.

4. Natural Language Processing

Natural Language Processing (NLP) is the area of AI concerned with computers working with human language. Common tasks include text classification, sentiment analysis, translation, summarization, question answering, and named entity recognition. Large Language Models (LLMs) learn patterns from large collections of text and can generate, summarize, and answer questions, but their outputs may be incorrect.

5. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation combines information retrieval with text generation. A RAG system searches a collection of documents for relevant passages and provides them as context to a language model. The model then generates an answer based on the retrieved information rather than relying only on its training.

6. The RAG Workflow

A basic RAG workflow has six stages: (1) load documents, (2) split long documents into smaller chunks, (3) convert each chunk into a numerical embedding, (4) store embeddings in a vector database, (5) retrieve chunks similar to a user's question, and (6) generate an answer using the retrieved context. Chunk size and overlap affect how much context is available during retrieval.

7. Embeddings and Vector Databases

An embedding is a list of numbers that represents features or meaning in text. Texts with related meanings often have embeddings that are close under a similarity measure such as cosine similarity. A vector database stores embeddings together with original text and metadata. When a question arrives, the system embeds it, compares it with stored vectors, and returns relevant records. ChromaDB is one example of a local vector database.

8. Responsible Use of AI

AI systems can reflect errors or biases in their data. Responsible development includes protecting private information, checking data quality, evaluating model behaviour, explaining limitations, and keeping humans involved in high-impact decisions. A document Q&A; assistant should say when the source does not contain enough information, avoid inventing facts, and identify the pages used to form an answer.

9. Review Questions

1. What is the main difference between Artificial Intelligence and Machine Learning?
2. Name the three common types of Machine Learning and give an example of each.
3. What is an embedding, and why is it useful in semantic search?
4. List the main stages in a basic RAG workflow.
5. What role does a vector database such as ChromaDB play?
6. Why should a document Q&A; assistant provide source-page citations?